Peer Response

by Mansour Saeed Mubarak Hamdan Al Hamdani - Monday, 11 August 2025, 8:25 AM
Your analogy of ABS to an "insurgent outlook" works effectively for highlighting their decentralizing and adaptive qualities. Through emergent behavior, particularly relevant in complex systems such as financial markets and global supply chains; Heppenstall et al. (2021) and Ionescu et al. (2024) support your view. Your consideration of reacting in real time against strategic foresight adds a valuable layer beyond traditional distributed problem-solving perspectives, suggesting ABS could serve both short-term adaptation and long-term planning. The point about overfitting is well-taken; blind training to a dataset without accounting for real-world stochasticity can reduce system reliability. Cross-validation could involve multi-scenario simulations with stress-testing and continuous real-world calibration (North & Macal, 2010). The concern for human oversight is valid; while it can mitigate bias, it might also stifle emergent properties that make ABS innovative. Possible efficiency improvements include reduced decision delays, better resource utilization, and resilience against failure conditions. Your framing ensures analysis goes beyond technological optimization to governance, ethics, and robustness—essential considerations for ABS in high-risk industries.

References

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by Mansour Saeed Mubarak Hamdan Al Hamdani - Monday, 11 August 2025, 8:23 AM
The merit of operator-based agent systems (ABS) was analysed with a wise contrast to centralized systems. The positive advantages include flexibility, fault tolerance, and decentralized decision-making, supported extensively by Axtell and Farmer (2025) and Cincotti et al. (2022). The observation relating to AI upheaval and independent agent adaptation through deep learning fits well, given the fast-moving nature of fields like supply chain and healthcare, where environmental variables transform quickly. The reference to reactive vs. deliberative or hybrid models provides useful context for integration avenues, although the question of multi-agent cooperation remains significant (Cincotti et al, 2022). The thought that simulations of organizational behavior are important is accepted, but as you fairly observe, prioritizing the good of the whole can be tricky; especially under competition. In real-time applications, conflict resolution methods may have to integrate adaptive algorithms with predefined coordination mechanisms to avoid deadlock or counterproductive rivalry. On the other hand, the hybrid approach could benefit industries such as logistics, energy distribution, smart manufacturing, and disaster response by merging decentralized adaptability with centralized supervision stability. Overall, your analysis addresses both the potentials and unresolved operational challenges concerning ABS.

References

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Cincotti, S., Raberto, M., Teglio, A. & Vighi, A. (2022) ‘Agent-based models for economic policy design: From simulations to implementations’, *Journal of Economic Behavior & Organization*, 198, pp. 336–356.

Peer Response

by Mansour Saeed Mubarak Hamdan Al Hamdani - Monday, 11 August 2025, 8:17 AM

Your association of ABS with decentralized company functioning is well-grounded, contrasting with centralized models and supported by Ding et al. (2023) and Sayyed-Alikhani et al. (2021). The emphasis on autonomy for real-time decision-making in supply chains is convincing, especially as it links to improved decision speed and responsiveness. Your recognition of ABS's potential for optimizing resource allocation; reinforced by Mueller-Bloch et al. (2024) in decentralized blockchain governance; broadens the discussion to governance and infrastructure design. Your reference to Aftabi et al. (2025) on mitigating threats in adaptive networks is valuable, as such systems can suffer cascading failures if coordination fails. Your caution on overdependence on agents is valid, as flawed decision rules or biased data can amplify systemic risks. Ethical adaptability could involve transparent decision protocols, continuous auditing, and hybrid human; AI oversight. Your suggestion on customer service automation adds another relevant application: ABS could enable personalized, real-time service delivery via adaptive virtual agents, while coordinating across numerous back-end systems, enhancing scalability and responsiveness in high-volume environments.

References

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